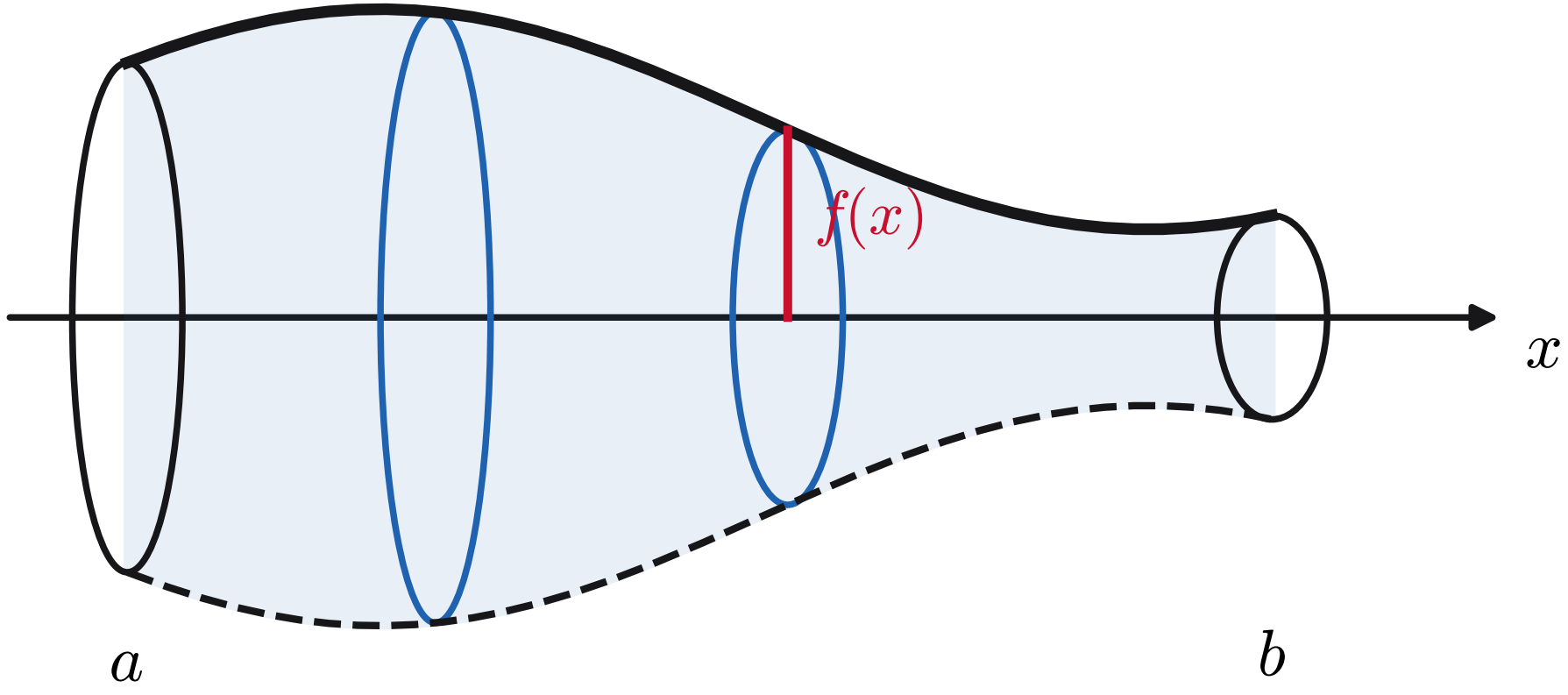$$V = \pi \int_a^b [f(x)]^2 dx$$